Final Year Project — Full Reference Document (v2, Revised/Staged Scope)

High-Precision Multi-Temporal Surface Area Analysis of Hyderabad's Urban Lakes Using Deep Learning

Compiled from project planning discussion, revised after guide feedback — keep this as your master reference.

> **v2 change log:** Restructured around a staged core/optional scope per guide feedback. Novelty reframed away from "transformer comparison" and toward "high-precision multi-temporal quantification of urban lake surface area change / encroachment." Grad-CAM and transformer comparison demoted from core objectives to optional add-ons. Difficulty revised upward (labeling flagged as the hardest component). Added a prioritized order for optional extensions, and a new lowest-priority stretch goal (shoreline change classification).

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1. One-Line Summary

Using AI (deep learning) to measure exactly how much Hyderabad's lakes/reservoirs (Hussain Sagar, Osman Sagar, Himayat Sagar) have shrunk over the past ~10 years, using satellite images — since no prior study has done this specific analysis for these lakes. FTL-based "encroachment" framing is used if boundary data can be obtained; otherwise the project reports on surface area change directly.

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2. Project Title

High-Precision Multi-Temporal Quantification of Urban Lake Surface Area Change in Hyderabad Using Deep Learning-Based Segmentation of Sentinel-2 Imagery

(Subtitle/extended version, if FTL data is obtained: "…with Full Tank Level-Based Encroachment Analysis")

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3. Abstract

Urban lakes play a critical role in flood control, groundwater recharge, and local ecology, yet rapid urbanization has led to significant shrinkage and degradation of these water bodies in many Indian cities. Hyderabad's iconic lakes — Hussain Sagar, Osman Sagar, and Himayat Sagar — have reportedly lost substantial surface area over recent decades due to encroachment and sewage inflow, but this shrinkage has not yet been quantified using a rigorous, deep-learning-based remote sensing pipeline. This project proposes a semantic segmentation approach using a U-Net convolutional neural network to automatically detect water body boundaries from multi-temporal Sentinel-2 satellite imagery (2016–2025). Using

NDWI/MNDWI-derived and manually verified ground-truth masks, the model is trained to distinguish water from non-water pixels with high spatial precision, including handling ambiguous cases such as algae/moss-covered water surfaces. The resulting per-year segmentation masks are used to compute surface area trends over time, providing a data-driven, year-by-year quantification of lake surface area change. Where official Full Tank Level (FTL) boundary data can be obtained, this quantification is extended into an encroachment analysis, distinguishing normal seasonal fluctuation from permanent loss of water-spread area. The core contribution of this work is the precision and rigor of the multi-temporal measurement pipeline itself, rather than any single architectural novelty. As optional extensions, Grad-CAM is applied as a qualitative interpretability check on the model's boundary predictions, and a transformer-CNN hybrid is evaluated against the U-Net baseline as a secondary comparison. This work aims to fill a specific gap in the literature: while deep-learning-based water body detection has been studied extensively across Indian and global datasets, no prior study applies this methodology specifically to Hyderabad's urban lake system. The findings are intended to support local water body monitoring efforts and contribute quantitative evidence to ongoing lake restoration discussions.

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4. Introduction

Water bodies in urban environments serve essential ecological and civic functions, including flood mitigation, groundwater recharge, temperature regulation, and recreational and cultural value. In rapidly urbanizing cities like Hyderabad, however, lakes have come under sustained pressure from encroachment, untreated sewage discharge, and land-use change. Hussain Sagar, one of Hyderabad's most prominent lakes, has been reported to have shrunk by approximately 40% over the past three decades, primarily due to sewage-fed drains and unregulated urban expansion around its periphery. Despite this well-documented ecological concern, no study to date has applied a deep-learning-based remote sensing pipeline to systematically quantify this shrinkage using satellite imagery.

Recent advances in deep learning have substantially improved the accuracy and scalability of water body detection from satellite imagery. Architectures such as U-Net and, more recently, transformer-CNN hybrids (e.g., WaterFormer) have demonstrated strong performance on water segmentation benchmarks, while transformer-based siamese networks like ChangeFormer have set a precedent for detecting land-cover change between time-stamped satellite images. Separately, growing interest in explainable AI (XAI) for remote sensing — using techniques such as Grad-CAM — has enabled researchers to interpret which spatial and spectral features drive a model's predictions, adding transparency to otherwise black-box deep learning systems.

While extensive prior work exists on deep-learning-based water body detection across various Indian and global datasets — including Faster R-CNN-based extraction, ResNet-based segmentation, and GeoAI approaches using ISRO's Cartosat-3 imagery — none of these studies focus specifically on Hyderabad's urban lake system. Similarly, existing remote sensing studies on Hussain Sagar have concentrated on water quality assessment (e.g., spectral reflectance and chromaticity analysis during the COVID-19 lockdown) rather than deep-learning-based extent detection or multi-temporal change quantification.

This project addresses that gap by applying a U-Net-based semantic segmentation model to multi-temporal Sentinel-2 imagery of Hyderabad's three major lakes, generating a quantified, year-by-year record of surface area change from 2016 to 2025. The primary contribution is the rigor and precision of this multi-temporal quantification pipeline for a previously unstudied local context, rather than a novel model architecture. Optional extensions — an explainability layer using Grad-CAM, a comparative transformer-CNN hybrid, and (data permitting) an FTL-based encroachment analysis — are pursued only after this core pipeline is validated. The resulting pipeline and findings aim to provide a locally relevant, technically rigorous contribution to Hyderabad's ongoing lake conservation and restoration efforts, while also serving as a reproducible template for similar urban lake monitoring studies elsewhere in India.

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5. Staged Scope — Core vs. Optional (Post-Guide-Review Structure)

This is the most important section to keep front-of-mind. Per your guide's feedback, the project is deliberately staged so the core is guaranteed-achievable, and everything else is added only if time/data allow — in priority order.

Core Objectives (must complete — this alone is a valid, defensible project)

1. Build a robust U-Net model to segment water bodies from Sentinel-2 imagery (handling algae/turbidity ambiguity via MNDWI + multi-band input).
2. Measure year-by-year lake surface area (2016–2025) for Hussain Sagar, Osman Sagar, Himayat Sagar.
3. Analyze and report surface area/shrinkage trends for the three lakes.

Optional Extensions (attempt only after the core is complete and validated, in this priority order)

1. **FTL-based encroachment analysis** — reframe surface area change against the notified Full Tank Level boundary, if shapefiles/boundary data can be obtained. Highest-priority add-on because it most strengthens the paper's rigor and real-world relevance — but explicitly **not guaranteed**, since the data source is unconfirmed (see Section 13).
2. **Grad-CAM explainability** — qualitative interpretability visualizations on the model's boundary predictions. Treated as "additional qualitative analysis," not a main objective or contribution.
3. **Transformer-CNN hybrid comparison** (e.g., vs. WaterFormer-style architecture) — secondary technical comparison, not the project's novelty claim. Novelty comes from the rigor and locality of the measurement pipeline, not from comparing architectures.
4. **Shoreline change classification** (lowest priority / only if significant time remains) — classifying shoreline segments into categories such as stable, seasonal fluctuation, permanent encroachment, and new construction nearby. This is a genuinely valuable extension that pushes the project from segmentation toward environmental monitoring, but it is effectively a second classification task layered on top of segmentation, requiring its own labels/logic — attempt only with real time to spare, and never let it displace the core objectives.

Why this structure matters: the transformer comparison and Grad-CAM were originally discussed as fairly central pieces of the project; guide feedback correctly identified that neither should carry the paper's novelty claim, and that over-scoping around them risks the whole project if time runs short. The core (objectives 1-3 above) is what you present and defend even in a worst case; the optional layers are what elevate it toward a stronger publication if things go smoothly.

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6. Important Conceptual Note: These Are Artificial Reservoirs, Not Natural Lakes

- **Hussain Sagar** — built in 1562, a dammed tank
- **Osman Sagar (Gandipet)** and **Himayat Sagar** — built in the 1920s as drinking water reservoirs

Why this matters: for artificial reservoirs, there's a precise, engineering-defined concept called **Full Tank Level (FTL)** — the maximum design water spread area. This is a better and more defensible baseline than a vague "historical extent," because it lets you separate: - **Normal seasonal fluctuation** (water level rising/falling with monsoon/summer — not real shrinkage) - **Permanent encroachment** (buildings/roads/slums built inside the FTL boundary that reduce capacity permanently — the real, policy-relevant story)

Refined research question: > "How much of Hussain Sagar/Osman Sagar/Himayat Sagar's officially notified Full Tank Level area is now permanently encroached upon, as detected across satellite imagery from 2016–2025 — distinguishing this from normal seasonal water level variation?"

Data needed for this: notified FTL boundary/shapefiles — potentially available via Telangana Irrigation Department, HMDA, or RTI/court filings related to Hyderabad lake encroachment litigation. (Not yet searched/confirmed — flagged as a to-do. **This is the single biggest risk to the "encroachment" framing** — see Section 11 and Section 5's staged scope. If this data cannot be obtained, the project reports on surface area change directly, without weakening the core deliverable.)

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7. Can Satellites Actually Detect This?

Yes — water has a very distinct spectral signature (strongly absorbs near-infrared light, unlike land/vegetation/buildings). This is why NDWI-type indices work.

Lake sizes (all comfortably large enough for Sentinel-2's 10m resolution): | Lake | Surface Area (at FTL) | |—|—| | Hussain Sagar | ~5.7 km² | | Osman Sagar (Gandipet) | ~46 km² | | Himayat Sagar | ~28 km² |

All three give tens of thousands of pixels each — more than enough detail for precise boundary tracing.

Important distinction — Area vs. Level: - Satellite optical imagery (Sentinel-2) reliably detects **surface area/extent change** — even a shoreline shift of ~20-30m across a stretch is detectable - It does **NOT** directly measure **water depth/level in meters** — that requires satellite altimetry (e.g., Sentinel-6, ICESat-2) or bathymetry/DEM data, which is a separate, more specialized technique and overkill for

this project - **Project scope decision:** stick to **surface area change** as your measured variable (the standard, accepted approach in all the base papers). You can reasonably infer "reduced area at the same time of year = reduced storage/level, assuming similar rainfall" as a defensible statement, without claiming direct depth measurement.

Usability check — all three lakes are good candidates, no swaps needed: | Lake | Usability | |—|—| | Hussain Sagar | ✅ Good — large enough; some algae/pollution noise (manageable) | | Osman Sagar | ✅ Excellent — large, clean, minimal urban interference | | Himayat Sagar | ✅ Excellent — same reasoning |

Practical scheduling note: Hyderabad's dry season (~Dec–May) gives the best cloud-free imagery window. Pick a **consistent month each year** (e.g., always March or April) so area comparisons are fair (dry-season to dry-season, not accidentally dry-season vs. monsoon-swollen).

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8. The Algae/Moss Problem (Water vs. Land Confusion)

The problem: algae/moss on water surfaces reflects light similarly to land vegetation (high NIR reflectance, like plants) — the opposite of clean water's signature (strong NIR absorption). Plain RGB or basic NDWI can genuinely misclassify algae-covered water as land.

How it's solved (layered approach):

1. **Use more spectral bands, not just RGB** — Sentinel-2 has 13 bands including SWIR (shortwave infrared), which behaves differently over algae-covered water vs. dry land/dense vegetation, even when NIR alone is ambiguous.

2. **MNDWI instead of (or alongside) plain NDWI** — Modified NDWI uses **Green/SWIR** instead of Green/NIR, specifically designed to be more robust against vegetation/algae confusion. This is the standard fix used in water-extraction literature for exactly this problem.

3. **Combine multiple indices** — NDWI + MNDWI + NDVI (vegetation index) + optionally AWEI (Automated Water Extraction Index). Algae-covered water shows a specific *mismatch* pattern across these indices that pure land doesn't show — this mismatch itself becomes a useful signal.

4. **This is where the deep learning model earns its keep** — a simple threshold-based index is fooled by algae, but a U-Net fed multiple spectral bands (not just one NDWI number) can learn the combined multi-band pattern *plus* spatial context (algae patches have distinct texture/shape and sit within the known lake boundary region). This is a legitimate, literature-backed justification for why DL beats simple index thresholding — not just a buzzword upgrade.

5. **Practical step:** during manual QGIS verification (Phase 2 of the project), specifically check and correct greenish/algae patches inside Hussain Sagar (known for algae blooms from sewage-driven eutrophication). This gives the model good training examples so it learns to handle the ambiguity.

Report-ready justification line: *"This project uses multi-band spectral input, MNDWI-assisted labeling, and CNN spatial context specifically because single-index approaches (plain NDWI) are known*

to misclassify algae-covered water as land — a well-documented limitation in remote sensing literature.”

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9. Tech Stack (Full List)

Data & Access - Google Earth Engine (GEE) — primary imagery source/access, Python API (`earthengine-api`) - Sentinel-2 SR (`COPERNICUS/S2_SR_HARMONIZED`) — 10m resolution, main dataset - Copernicus Browser (https://browser.dataspace.copernicus.eu/) — manual browsing/cloud-cover check - USGS EarthExplorer — Landsat, for pre-2016 data if needed - ISRO Bhuvan (https://bhuvan.nrsc.gov.in/) — optional Indian-satellite cross-check - QGIS — manual verification/correction of masks, ground-truth polygon drawing

Core ML/DL - Python 3.10+, PyTorch - segmentation-models-pytorch (U-Net w/ pretrained ResNet34 encoder) - Albumentations (augmentation) - rasterio, geopandas (geospatial I/O) - numpy, opencv

Explainability - pytorch-grad-cam - captum (backup/alternative)

Compute & Tracking - Google Colab (free/Pro) - Weights & Biases (W&B)

Visualization/Reporting - Matplotlib/Seaborn - Folium / Kepler.gl (interactive before/after maps) - Overleaf (LaTeX) or Word for final report

Optional benchmark/sanity-check dataset - Kaggle “Satellite Images of Water Bodies” — test pipeline before running on real Hyderabad data

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10. Difficulty Breakdown (Revised After Guide Review)

Your guide rated this slightly harder than the original estimate, mainly because manual annotation/labeling is more time-consuming and judgment-heavy than it first appears. Revised ratings adopted below:

Component	Difficulty
Learning satellite imagery basics	⋆⋆⋆
Data collection (GEE)	⋆⋆
Manual annotation/labeling (incl. algae correction)	⋆⋆⋆⋆ — hardest core component
U-Net baseline	⋆⋆
Multi-temporal / time-series analysis	⋆⋆⋆
Report/paper writing	⋆⋆⋆
Transformer hybrid (optional)	⋆⋆⋆⋆
Grad-CAM explainability (optional)	⋆⋆⋆
Shoreline change classification (optional, lowest priority)	⋆⋆⋆⋆ (a second classification task on top of segmentation)

Overall project difficulty: ⋆⋆⋆⋆ (~3.8/5) — still feasible solo, but harder than it first appears. Mitigate the labeling bottleneck by capping manual annotation at roughly 300–500 patches and leaning

on transfer learning (pretrained encoders) rather than attempting to label thousands of patches from scratch.

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11. Timeline (16 weeks / ~4 months core, buffer to 5-6 months)

Weeks	Phase
1–3	Literature review, proposal
4–6	Data collection & preprocessing (incl. cloud masking, NDWI/MNDWI, manual QGIS verification incl. algae correction — **cap labeled patches at 300–500**)
7–9	U-Net baseline training
10–11	**Core deliverable:** surface area / shrinkage trend analysis for the three lakes
12–13	*(Optional, priority order)* FTL-based encroachment reframe (if data obtained) → Grad-CAM → transformer comparison
14–15	*(Optional, only if time remains)* Shoreline change classification
15–16	Report writing, defense prep

Discipline point: weeks 10–11 already produce a complete, defensible project. Everything from week 12 onward is upside, not requirement — don't let an optional layer eat into report-writing time.

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12. Risk Points to Watch

- **FTL boundary data may not be obtainable — this is the biggest risk to the proposal.** The whole "encroachment" framing depends on official Full Tank Level shapefiles/boundaries, which are not yet confirmed to be publicly available. Mitigation: the core project is scoped around surface area change, which needs no external boundary data at all; FTL/encroachment is added only if the data comes through, never assumed as guaranteed.
- Monsoon cloud cover — mitigated by consistent dry-season month selection
- NDWI mislabeling polluted/algae-covered water — mitigated by MNDWI + manual QGIS correction + DL's multi-band learning
- Small dataset size (only 2-3 lakes) — mitigated by transfer learning (pretrained encoders) + heavy augmentation
- Manual annotation is the most time-consuming step — mitigated by capping labeled patches at 300–500 rather than attempting thousands
- Scope creep — lock in the three core objectives (Section 5) before touching any optional extension, and follow the stated priority order rather than working on whichever is most interesting

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13. Novelty Justification (for defense/report — Revised)

The novelty claim is the rigor and locality of the multi-temporal measurement pipeline — not any single model architecture. Do not lead with “we compare U-Net to a transformer” as the novelty; that framing is weak, since architecture comparisons are extremely common in the literature. Instead lead with:

> “High-precision, multi-temporal quantification of Hyderabad urban lake surface area change (and, where data permits, FTL-based encroachment) using deep learning — a locally-specific analysis that does not yet exist in the literature.”

No existing study combines: 1. Deep learning-based segmentation 2. Applied specifically to Hyderabad’s lake system 3. Rigorous, multi-temporal, year-by-year quantification (2016–2025) 4. (Optional layer) Compared against notified FTL boundaries — encroachment framing, not just vague “shrinkage” 5. (Optional layer) With explainability (Grad-CAM) as qualitative support, and/or a transformer-hybrid as secondary comparison

Existing related work (confirmed via search) is either: generic-region DL water detection (Faster R-CNN, ResNet, Cartosat-3/GeoAI), or Hussain-Sagar-specific but water-quality-focused (not DL/extent-based).

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14. Publication Chance Assessment (Per Guide Review)

If executed well, following the staged core-first approach:

Venue	Chance
Internal college publication	⋆⋆⋆⋆⋆
National conference	⋆⋆⋆⋆☆
IEEE conference	⋆⋆⋆☆
SCI journal	⋆⋆☆

This is considered a strong outcome for a solo B.Tech final year project.

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15. Guide Feedback Summary (Received and Adopted)

Overall verdict from guide review:

Category	Rating
Novelty	⋆⋆⋆⋆☆
Technical Depth	⋆⋆⋆⋆☆
Difficulty	⋆⋆⋆⋆☆
Publication Potential	⋆⋆⋆⋆☆
Feasible Solo?	✓ Yes, with discipline

Key changes adopted from this feedback (all reflected above): - Novelty reframed away from “transformer comparison” toward “high-precision multi-temporal quantification / encroachment analysis” (Section 13) - Grad-CAM demoted to qualitative/optional, not a core objective (Section 5, 8) - FTL boundary dependency explicitly flagged as the top project risk, with surface-area-only as the guaranteed fallback (Sections 5, 6, 12) - Manual annotation explicitly capped at

300–500 patches (Sections 10, 11, 12) - Difficulty rating revised upward to ⋆⋆⋆⋆ (~3.8/5) - Added a new lowest-priority stretch goal: shoreline change classification into stable / seasonal fluctuation / permanent encroachment / new construction categories (Section 5) — noted as valuable but a genuine scope increase, not a “free” addition, so it sits below even the transformer comparison in priority.

Guide’s overall recommendation: favor this project over the methane emission alternative — same environmental monitoring domain, but significantly more achievable for a solo undergraduate while retaining a meaningful, defensible research contribution.

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16. Open To-Dos / Next Steps Not Yet Done

- ☐ Search for actual FTL notification shapefiles/boundaries (Telangana Irrigation Dept / HMDA / court filings)
- ☐ Write actual Google Earth Engine Python script to pull + export Sentinel-2 imagery for the 3 lakes
- ☐ Set up GEE account + authenticate Python API
- ☐ Draft separate “Problem Statement” and “Objectives” sections for proposal format (if guide requires them distinctly from abstract/intro)
- ☐ Decide, before locking the proposal, whether FTL data availability is confirmed enough to include in the title/abstract, or whether to keep the title FTL-agnostic (current recommendation: keep title FTL-agnostic, per Section 2)